Biotechnology Design

TSA State Conference - Newark, Delaware
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Problem: Injuries Resulting In Loss of Limbs

The 2025 theme for Biotechnology Design is tissue engineering. Tissue engineering has various branches such as severe burns, organ issue, transplant rejection, bad plastic surgeries, and limb injuries. After reviewing these numerous problems, my team and I peaked an interest in limb loss and injuries. A majority of us had watched Grey's Anatomy and orthopedic surgery interested in our STEM driven minds. There are numerous solutions to limb loss such as bone grafting, regrowth of bones, and scaffolding to name a few.

Many individuals have fallen victim to limb loss, including veterans and newborn babies. The point is, there are many individuals who can benefit from medical research in the solutions of limb loss. Every year more than five hundred thousand Americans experience limb loss or are born with a disability that deprives them of a limb. Limb loss can transpire from war injuries, birth defects, trauma, infection, and more.

The impact of sustainable solutions of limb loss would be extensive for communities; covering the environment, economy, and society. Concerning scaffolding, parts could be 3D printed and filament of 3D printers are reused plastic. Scaffolding could also use seaweed, clay, and bioglass; all natural materials. Bone regrowth through bone remodeling using numerous cells to model a bone. Bone grafting also includes environmentally friendly aspects, such as auto grafting. Auto grafting takes parts of the hip bone or another part of the body to regenerate a piece of the diseased bone.

However, natural or synthetic scaffolding, our third solution, is not economically friendly. A surgery can cost anywhere from fifty thousand to one hundred and fifty thousand. Due to the surgery being in its early stages and insurance will not be able to fully cover the procedure. Also, our second solution, regrowing bones through bone remodeling, cost several thousand dollars for a procedure due to how new and experimental the treatment is. But, our chosen solution, bone grafting: solution one, can cost anywhere from two hundred to three thousand dollars. Society wise, bone grafting is more sustainable than natural or synthetic scaffolding. Scaffolding is only a temporary solution and is detrimental to individuals to participate in innocuous activities.

Solution One: Bone Grafting - Chosen Solution

Bone grafting is the process of replacing damaged or diseased bone with healthy bone. Because bone grafting works in almost any area in the body it can be used to treat many different bone problems/bone diseases such as bone fractures, bone cancer, spinal infusion, and dental implants. Bone grafts can promote the growth of healthy bone and can also help repair bones that are not likely to heal properly and can help strengthen damaged bones. There are some risks to a bone grafting procedure which include bleeding, infections,failure of the graft, decreased mobility, and chronic pain. The recovery time can vary between people due to age, physical health, and the type of surgery that was needed.

There are 3 main types of bone grafting: Auto grafting, Allografting, and Synthetic Bone grafting. The first is autografting. Autografting is when the bone needed for the graft is taken from somewhere in your own body, usually it is taken from the top of your hip bone. Because the bone is from your own body there is no risk of getting a disease but a drawback is that there is a limited amount of bone tissue they can take. There is also an increased chance of a successful fusion.

Second is Allo Grafting where the bone that they use to replace the injured bone is gotten from a person who has died. Because it is from a different person there are many regulations to make sure the bone is sterilized and safe for the person receiving it. However, there is still a risk of infections and other complications. The most common bones used in Allo grafting are harvested from the leg bones such as Tibia, fibula, and femur; they also use ribs.

Lastly, synthetic bone grafts are grafts that can be made of many different materials and can be a good substitute for other grafting methods by removing some of the cons of the other method like the chance of an infection. While other methods are more limited due to them requiring donors and donor sites, synthetic grafts could be made out of many different materials, even some of them having benefits like promoting bone fusion and more.

Solution Two: Regrowing Bones Through Bone Remodeling

The purpose of regrowing bones from stem cells is to improve individuals' skeletal frame and to eventually their way of life. Biotech specialist Nina Tandon explains what is needed from the patient to recreate a bone in *"How To Regrow Your Own Bones"* published by Scientific American Magazine. "We take two things from the patient. One is a sample of their fat tissue, from which we extract stem cells. The other is a CT scan." The CT scan is used to form the perfect scaffold shape. The scaffold can is either created by using protein and collagen from animal bones, or synthetic materials. Similar to the third solution we covered: Natural and Synthetic Scaffolding. The cells from the patient are then infused into the model of a scaffold. In a course of three weeks the cells form into a bone for the patient.

The cells sampled to recreate a bone are: osteoblasts, osteocytes, bone lining cells, and osteoclasts. Osteoblasts are known as bone-forming cells because they produce and release collagen and proteoglycans that create the framework of bone tissue. Osteocytes are cells that live within the bone matrix. The main purpose of these cells is to remove damaged organelles and macromolecules using quality-control pathways. Both osteoblasts and osteocytes induce bone remodeling since they produce osteoclastogenic factors. Bone lining cells cover the surface of bones and break off from osteoblasts. These cells form a barrier between osteoclasts and bone matrix in order to prevent interaction amongst the cells. Osteoclasts are the forth cell that contributes to regrowing a bone. Its function is to break down old or damaged bone tissue in order to create room for new bone tissue.

The disadvantage of using osteoblasts, osteocytes, bone lining cells, and osteoclasts to regrow bones would be the inadequate amount of blood supply. Due to this, the bone tissue would not be able to fully develop properly. Also, solely using bone cells would stunt the development of new blood vessels in the regeneration site. Depending on the complexity the cost of a procedure could be several thousand dollars due to how new and experimental the treatment is. The procedure requires advanced surgical skills and specialized equipment. As well as, a complex process of isolating and preparing the required bone cells of the patient's tissue. In conclusion, regrowing

Solution Three: Natural and Synthetic Scaffolding In Bone Regeneration

Natural and synthetic scaffolding is a method for individuals who are in need of a temporary replacement of bone. The purpose of scaffolding is to recreate the aspect of the cell microenvironment. New tissue can be regenerated when cell proliferation, migration, differentiation, and synthesis of a new extracellular matrix occur. This method, an artificially produced bone, allows dependability and minimal risk of disease. There are numerous materials that can be used to create a scaffolding. To form a natural scaffolding, materials such as collagen, alginate (seaweed), chitosan (shellfish), hyaluronic acid, and fibrin (blood). To form a synthetic scaffold, polylactic acid, polyethylene glycol, polycaprolactone, and polyglycolic acid is used.

Scaffolds can be formed based on parameters of pore size and shape, porosity and pore interconnectivity, wall morphology, and surface area which are key issues for cell seeding, migration, growth, mass transport, and tissue formation. There are many types of scaffolding fabrication techniques such as, 3D printing, rapid prototyping solvent-based techniques, solvent casting, and phase separation. One type of synthetic scaffolding is biomaterial scaffolds, it can be made with or without cells. Biomaterial scaffolds mimic the extracellular matrix that normally maintain space, support cells, and organize cells into tissue.

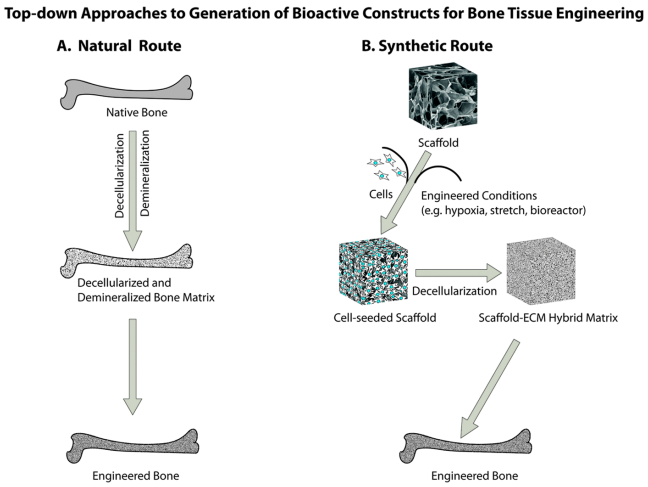
The main issue with this solution is the durability of the bioglass. Bioglass can be a temporary fix and sustainable for individuals who don't plan on participating in innocuous activities. For individuals who partake in precarious activities, it is recommended to pursue a more sustainable solution. *"A Review of the Application of Natural and Synthetic Scaffolds in Bone Regeneration"* written by Pei-Chun Wong explains another disadvantage of natural and synthetic scaffolding. "The lack of sufficient oxygen and nutrient supply and the accumulation of metabolic waste products limit cell growth and form irregular cell density within the scaffold. This problem can be solved by developing a bioreactor system which provides a dynamic environment for cell-load scaffold culture."

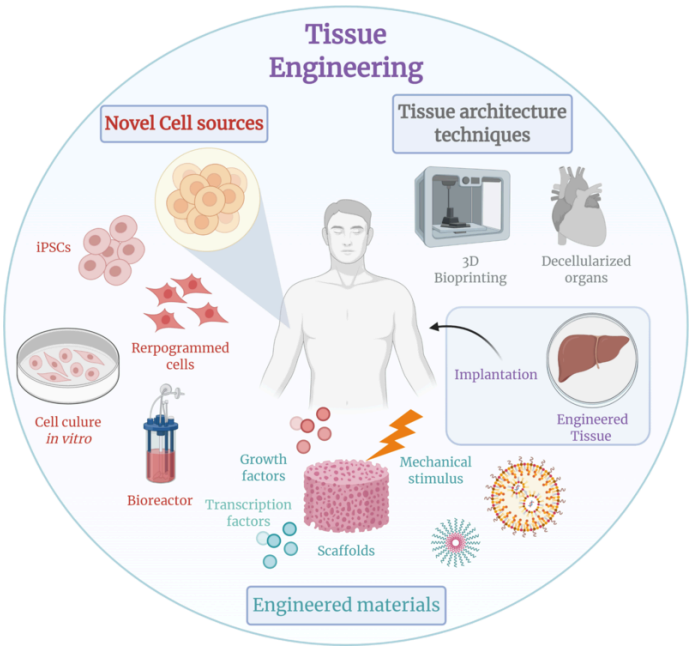
Real-Life Applications

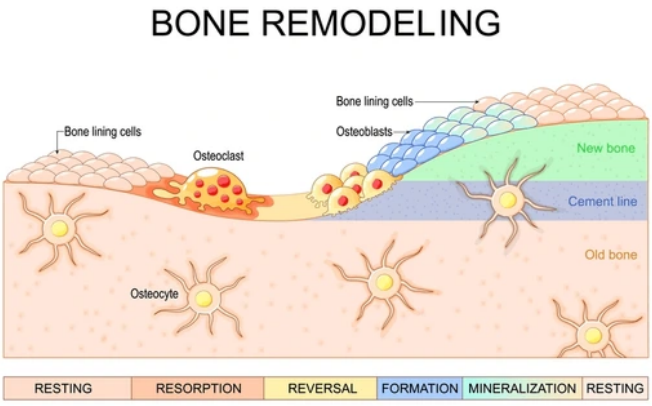
A 5-year-old boy was taken to the Children's Hospital of Philadelphia due to having pain and trouble using his shoulder in his right arm after a minor trauma. After an MRI and an image-guided biopsy were performed he was diagnosed with osteosarcoma. A team of oncologists had to figure out a plan of action that would not harm the proximal humerus growth plate and sustain the activities he would do in the future. They came up with a vascular fibular graft which would replace the bone with the tumor with bone from his fibula. After the surgery, he went through intense physical therapy and is now a healthy 6-year-old.

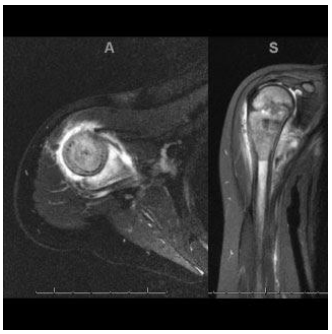
Six patients two with osteosarcoma and four with Ewing sarcoma underwent tumor retraction all using tabula strut autografts for reconstruction. A union was achieved in five patients; the last one did not reach a union fully only at the distal end. The average time for a union between the proximal junction and the distal junction was four to eight point eight months. Three out of six developed joint arthrodesis two in the knee and one in the ankle. Two had to get leg lengthening surgery due to their legs being significantly different sizes. One developed a foot drop. Four patients lived with no further disease and two died due to the original diseases.

Supplemental Information









Multimedia Presentation

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Problem: Injuries Resulting In Loss of Limbs

- **Victims/Reasons of Limb Loss**
  - Veterans: War injuries
  - New borns: Birth defects
  - Any Individual: Trama
- **How Economically Friendly Is It?**
  - Solution two: Regrowing bones through bone remodeling
  - Costs several thousand of dollars for a procedure
  - New and experimental solution
  - Requires advanced surgical skills and specialized equipment
  - Complex process isolating patient bone cells

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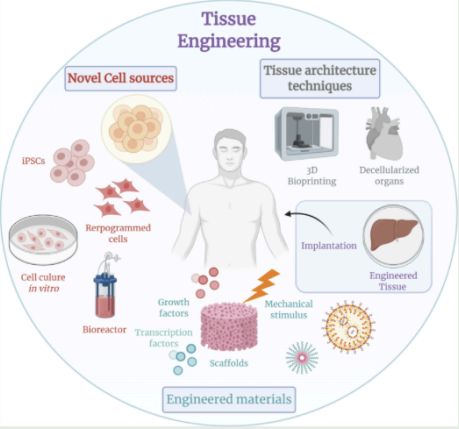
Problem: Injuries Resulting In Loss of Limbs

- **How Economically Friendly Is It?**
  - Solution three: Natural and synthetic scaffolding
  - Costs anywhere from $50,000 to one $150,000
  - Survey in early stages
  - Insurance will not fully cover procedure
  - Not permanent sustainable solution
- **How Economically Friendly Is It?**
  - Solution two: Bone grafting - chosen solution
  - Costs between $200 and $3,000
  - More sustainable than other solutions



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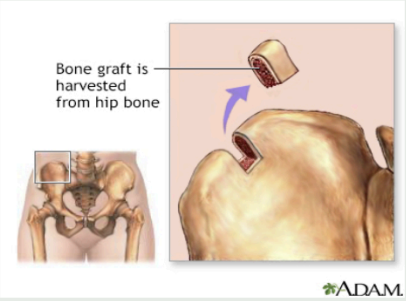
Solution 1: Bone grafting - Chosen Solution

- **Bone Grafting**
  - Performed anywhere in body (commonly used in knees, hips, and spine)
  - Treat numerous types of bone problems like bone cancer, bone fractures, and spinal infusions

- **Auto Grafting**
  - The graft is from the patient's body
  - Usually taken from the iliac crest (top of hip)
  - Greatest chance of successful fusion
  - No chance of a disease.



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Solution 1: Bone grafting - Chosen Solution

- **Allo Grafting**
  - A graft that is donated from someone who has died.
  - Bones used: Tibia, fibula, femur, and ribs
  - High chance of failure and disease

- **Synthetic Bone Grafting**
  - Made with numerous materials and is more readily available
  - Less cons than other methods (no chance of disease)
  - Chance of failure later due to certain of materials degrading or breaking

- **Risks**
  - Infections, bleeding, decreased mobility, chronic pain, and failure of graft

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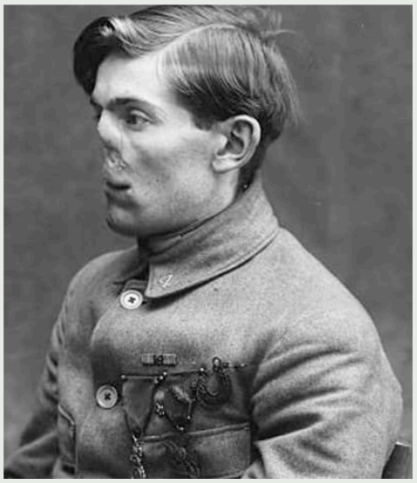
Solution #2: Regrowing Bones Through Bone Remodeling

- **Osteoblasts**
  - Bone forming cells
  - Produce and release collagen and proteoglycans
  - Creates framework of bone tissue
- **Osteocytes**
  - Cells that live in bone matrix
  - Removes damaged organelles and macromolecule
- Both cells induce bone remodeling since producing osteoclastogenic factors



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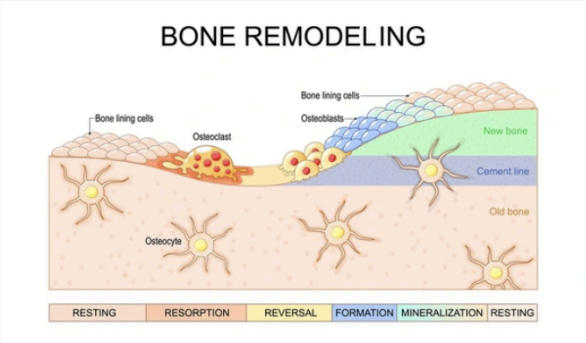
Solution #2: Regrowing Bones Through Bone Remodeling

- **Bone Lining Cells**
  - Covers surface of bones
  - Produced from osteoblasts
  - Barrier cell betw and bone matrix to prevent interaction
- **Osteoclasts**
  - Breaks down old and damaged bone tissue
  - Creates room for bone tissue



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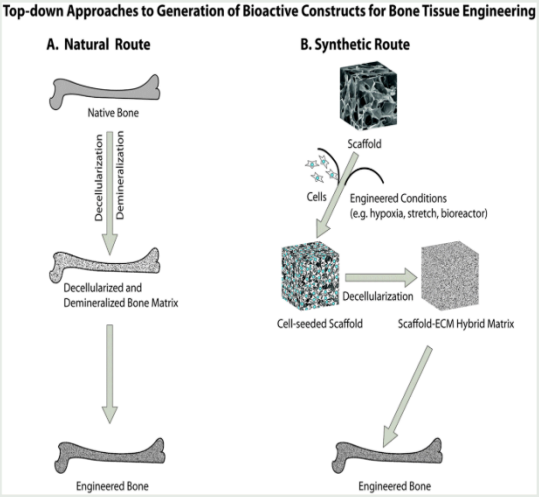
Solution #3: Natural & Synthetic Scaffolding

- **Natural Scaffolding Materials**
  - Collagen
  - Alginate (seaweed)
  - Chitosan (shellfish)
  - Hyaluronic acid
  - Fibrin (blood)

- **Synthetic Scaffolding Materials**
  - Polylactic acid
  - Polyethylene glycol
  - Polycaprolactone
  - Polyglycolic acid



Solution #3: Natural & Synthetic Scaffolding

- **Scaffolding Fabrication Techniques**
  - 3D printing
  - Rapid prototyping solvent-based
  - Solvent casting
  - Phase separation

- **Problems With Scaffolding**
  - Lack of oxygen (O2)
  - Lack of nutrient supply
  - Limited cell growth of metabolic waste products
  - Irregular cell density within the scaffold
  - Not durable due to bioglass
  - Chance of disease

Real life applications

- **Case Information**
  - 5-year-old boy
  - Children's Hospital of Philadelphia
  - Pain and trouble rotating right shoulder
  - Minor trauma
- **Diagnosis**
  - Had a MRI
  - Had an image-guided biopsy
  - Diagnosed with osteosarcoma
- **Treatment Process**
  - Oncologist needed plan to not harm the proximal humerus growth plate and detain patient from activities
  - Vascular fibular graft: Replaced disease bone with part of his fibula

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Real life applications

- **Case Information**
  - Total of 6 patients
  - Ages ranging from 6 to 15
- **Diagnosis**
  - 2 patients had osteosarcoma
  - 4 patients had ewing sarcoma
- **Treatment Process**
  - All patient got tumor resections
  - All 6 patients got tibial autografts
- **Complications Post Treatment**
  - 3 out of 6 developed joint arthrodesis (2 in knee and 1 in ankle)
  - 1 developed a foot drop
  - 2 had proportional different sized legs
  - 4 patient lived with no disease
  - 2 died from original disease

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Work Log

Resources

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